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Retrodiction of forest demography: Backward simulation with reverse matrix models

Nicolas Picard¹ | Jack Andrew Hansen² | Nancy Harris³ | Jingjing Liang²¹GIP Ecofor, Paris, France²Forest Advance Computing and Artificial Intelligence Lab (FACAI), Department of Forestry and Natural Resources, Purdue University, West Lafayette, Indiana, USA³World Resources Institute, Washington, District of Columbia, USA**Correspondence**

Nicolas Picard

Email: nicolas.picard.1991@polytechnique.org**Funding information**

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Handling Editor: Maria Pennino**Abstract**

1. Reconstructing past ecological population dynamics and demographic events is crucial for understanding the dynamics of ecological processes, evaluating the impact of environmental changes and making informed conservation decisions. In forest ecosystems, retrodiction (i.e. the backward projection of ecological populations) plays a pivotal role in understanding historical forest carbon levels and the factors that have influenced their variation over time, because forest demography is a major determinant of the amount of carbon stored in forest ecosystems. The persistent lack of quantitative methods has been a significant obstacle in retrodicting forest demography, especially in applications of a broad geographical scale. While there is a wealth of models for predicting future forest conditions, models that can project these conditions backward in time are scarce.
2. This study presents reverse matrix model (RMM), an innovative retrodiction modelling approach grounded in the principles of transition matrix models. RMM is designed to deduce past demographic characteristics of ecological populations using current data, making it one of the first models capable of projecting the fine-scale dynamics of forest demography into the past.
3. We assessed the retrodictive performance of RMM by fitting it to a dataset of a disturbed tropical rainforest in French Guiana in 2001–2023, then comparing the retrodictions to observations back to 1983 when the disturbance occurred. We further empirically evaluated the viability of retrodiction over a defined duration by inverting the density-dependent matrix model by Lin et al. (1996), which predicts the dynamics of northern hardwoods in the United States.
4. The case studies demonstrate significant potential for RMM application in various domains of forestry and conservation, including ecosystem management and conservation planning, global change impact assessment and biodiversity monitoring.

KEYWORDS

backcasting, backward projection, forest dynamics, matrix population model, retrodiction, transition matrix

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1 | INTRODUCTION

The backward projection of ecological populations, sometimes also referred to as retrodiction or backcasting, is the process of reconstructing past population dynamics and demographic events. Retrodiction is crucial for understanding the dynamics of ecological processes, evaluating the impact of environmental changes and making informed conservation decisions. Early applications of retrodiction were primarily in the field of population genetics. Kingman (1982) introduced the coalescent process, which utilizes the genealogical relationships among individuals to infer past population dynamics. Hudson (1990) and others (Gill et al., 2016) extended these ideas, applying the coalescent theory to infer past population sizes, migration rates and other demographic parameters.

In forest demography, which quantifies the structure and dynamics of tree populations within forest ecosystems, retrodiction can be used to infer historical forest conditions, such as the distribution of forest tree population by species and size. To this end, forest growth models, typically used to predict future forest conditions (Pretzsch, 2009), can be adapted to project forest conditions backward in time based on current forest conditions and historical data such as land use, fire history, or climate records (Bonnicksen & Stone, 1982). In climate change research, forest growth models can be used to understand how forests have responded to past climate variability and change (Barth et al., 2015). In forest conservation and restoration, backward projections can provide baseline or reference conditions to help restore ecosystems to their historical conditions, or understand how far current conditions have deviated from the natural state (Manning et al., 2006). These applications demonstrate the versatility of forest growth models not only in forecasting future scenarios but also in providing insights into the past, which is essential for informed forest management and conservation planning.

Forest demography is a major determinant of the amount of carbon stored in forest ecosystems (McDowell et al., 2020). Due to inconsistent data, methods and assumptions employed in existing forest carbon monitoring systems (Harris et al., 2021), large uncertainties in the magnitude and direction of the response of the terrestrial carbon cycle to climate change still remain unquantified (Pugh et al., 2020). Consequently, retrodiction plays a pivotal role in our comprehension of historical forest carbon levels and the factors that have influenced their variation over time. Understanding these dynamics is essential for reconstructing historical carbon storage benchmarks (Gasser et al., 2020; Magerl et al., 2019), which can inform predictive models and help in planning conservation and management strategies to mitigate climate change impacts.

Transition matrix models are widely used in many disciplines (Leslie, 1945; Lewis, 1942; Salguero-Gómez et al., 2015), especially in forestry where they have been employed to study the dynamics of almost all the forest types around the world (Liang & Picard, 2013). In the typology of forest growth models, matrix models stand between stand-level models and individual tree models (Porté & Bartelink, 2002). Stand-level models are useful for monospecific even-aged stands, but become inappropriate for mixed or uneven-aged stands because they do not consider within-stand variability

in individual size or species. In comparison, individual-based models predict the temporal changes of each individual tree on a stand, but due to their computing intensive nature, they have limited applications in large-scale studies. Matrix models provide a good compromise between computing requirements over large scale and capacity to deal with individual variability.

Matrix models have been applied to study almost all the aspects of forestry (Ohse et al., 2023). In forest ecology, matrix models have been used to study the demography of natural succession (e.g. Alvarez-Buylla, 1994) and the impacts of climate changes (Liang et al., 2011), and natural disturbances (Price & Bowman, 1994). In forest management, matrix models have been applied to evaluate forest carbon sequestration and biomass potential (Ingram, 2012; Liang, 2012), economic outcomes (Ingram & Buongiorno, 1996) and ecological impacts (Ma et al., 2016, 2020) of various management regimes, including the impact on the habitat of endangered wildlife species (Zhou et al., 2008).

The persistent lack of quantitative methods has been a significant obstacle in retrodicting forest demography, especially in applications of a broad geographical scale. While there is a wealth of models for predicting future forest conditions, models that can project these conditions backward in time are scarce. The mathematical methods that underpin matrix modelling pave the way for retrodicting matrix models. Matrix models are related to several mathematical fields, such as discrete-time finite-space Markov chains (Bruner & Moser, 1973; Caswell, 2001, Chap. 5) and partial differential equations (Caswell, 2001, Chap. 8; Takada & Hara, 1994). The retrodiction of Markov chains (Surace & Scandi, 2023) and the time-reversal of partial differential equations thus provide hints for retrodicting matrix models. However, matrix models do not entirely fall within the scope of these mathematical fields, so that none of them completely defines how to retrodict matrix models.

This study presents reverse matrix model (RMM), a retrodiction modelling approach grounded in the principles of transition matrix models. RMM is designed to deduce past demographic characteristics of ecological populations using current data, marking it one of the first models capable of projecting the fine-scale dynamics of forest demography into the past. Two related questions were addressed: (i) How can a RMM be fitted to data? (ii) Given a forward matrix model fitted to data, how can it be inverted using RMM or another inversion method? For question (i), we hypothesize that RMM has a retrodictive capacity similar to the predictive capacity of matrix models, as they are grounded in the same theoretical framework. For question (ii), we hypothesize that, because some information is lost in the forward direction, inversion is possible only within a bounded time frame. We assessed the retrodictive performance of RMM by fitting it to a dataset of a disturbed tropical rainforest in French Guiana in 2001–2023, then comparing the retrodictions to observations back to 1983 when the disturbance occurred. In addition to RMM, we also devised explicit methods to invert a matrix model from a forward to a backward recurrence relationship. We empirically evaluated the duration of time during which the backward dynamics followed the same trajectory as the forward dynamics by inverting the density-dependent matrix model by Lin et al. (1996), which predicts the dynamics of northern hardwoods in the United States.

2 | MATERIALS AND METHODS

2.1 | Matrix model

In a matrix model of forest dynamics, a population is represented by a vector $\mathbf{N}$ giving the abundance of individuals in K diameter classes at a given time t . Time is discrete. The dynamics is determined by the matrix recurrence relationship $\mathbf{O} = \mathbf{GN} + \mathbf{R}$, where $\mathbf{O}$ is the vector of tree abundances at the subsequent time $t + 1$, $\mathbf{G}$ is a $K \times K$ growth transition matrix, and $\mathbf{R}$ is a vector of length K whose first element gives the number of recruited individuals between times t and $t + 1$ and all other elements are null. By iterating the forward recurrence relationship that computes $\mathbf{O}$ from $\mathbf{N}$, future dynamics can be predicted. Between times t and $t + 1$, an individual may either move up from its diameter class to the next one, stay alive in its diameter class, or die. Hence, the growth transition matrix is written:

$$\mathbf{G} = \begin{bmatrix} q_1 & & & 0 \\ p_1 & q_2 & & \\ & \ddots & \ddots & \\ 0 & & p_{K-1} & q_K \end{bmatrix} \quad (1)$$

where q_i is the stasis rate in class i and p_i is the upgrowth rate from class i to $i + 1$. The mortality rate in class i is obtained as $m_i = 1 - p_i - q_i$.

A particular case is when the first element of $\mathbf{R}$ is a linear function of $\mathbf{N}$, that is written as ${}^t\mathbf{fN}$ where superscript t denotes the transpose and $\mathbf{f} = (f_1, \dots, f_K)$ is a vector of K fecundities. Most of the time, newly recruited trees cannot be related to their parent trees, so that the average fecundity $f = ({}^t\mathbf{1}_K\mathbf{R}) / ({}^t\mathbf{1}_K\mathbf{N})$ is the only estimable quantity. Then, the matrix relationship becomes: $\mathbf{O} = \mathbf{UN}$, where $\mathbf{U}$ is a Usher (1966, 1969) transition matrix:

$$\mathbf{U} = \begin{bmatrix} q_1 + f & f & \dots & f \\ p_1 & q_2 & & 0 \\ & \ddots & \ddots & \\ 0 & & p_{K-1} & q_K \end{bmatrix}$$

A matrix model predicts the transitions and per capita asexual or sexual contributions between the states of a life cycle

(Caswell, 2001). In forest dynamics, the states of the life cycle are diameter classes, and transitions are stasis, upgrowth and mortality (Figure 1). The trajectory of an individual along the life cycle can be described by a Markov chain. Given a dataset giving the state of every tree at time t and $t + 1$, the matrix model is calibrated by counting, for each diameter class i , the number M_i of trees that die, the number Q_i of trees that stay alive in diameter class i , and the number P_i of trees that move up from diameter class i to $i + 1$ (Figure 2). Transition rates are computed with respect to the number of living trees in each diameter class (i.e. excluding the number of dead trees). Thus, the forward transitions rates are: for class $i = 1, \dots, K - 1$, $m_i = M_i / N_i$, $q_i = Q_i / N_i$ and $p_i = P_i / N_i$, where $N_i = M_i + Q_i + P_i$ is the number of living trees in diameter class i at time t ; for the last class, $m_K = M_K / N_K$ and $q_K = Q_K / N_K$, where $N_K = M_K + Q_K$ is the number of living trees in diameter class K at time t .

Matrix models with fixed transition rates may not be sufficiently flexible to account for population dynamics (Caswell, 2001, Chap. 16; Roberts & Hruska, 1986). Transition matrix may actually vary between populations depending on their density. Density-dependent models where the transition rates are functions of $\mathbf{N}$ provide more realistic predictions when density-dependent processes such as competition affect vital rates (Favrichon, 1998; Lin et al., 1996). Density-dependent models are often fitted using a two-step process (Solomon et al., 1986). First, the transition rates for different populations at varying densities are estimated as described above. Second, these transition rates are regressed against population density.

2.2 | Reverse matrix model

A backward matrix model can be defined by reversing the direction of the transitions between the states of the life cycle (i.e. by reversing the direction of the arrows in Figure 1). Because the life cycle description of matrix models complies with a Markov chain, the retrodiction of Markov chains offers a *prima facie* way to develop RMMs. Let i and j be two states of the life cycle that are connected by an arrow, and let X_t be the state of an individual at time t . By definition, the transition rate from i to j in the matrix model is the probability $\Pr(X_{t+1} = j | X_t = i)$. Retrodicting the matrix model consists in computing the backward transition rate from j to i , that is the probability $\Pr(X_t = i | X_{t+1} = j)$. The retrodiction of

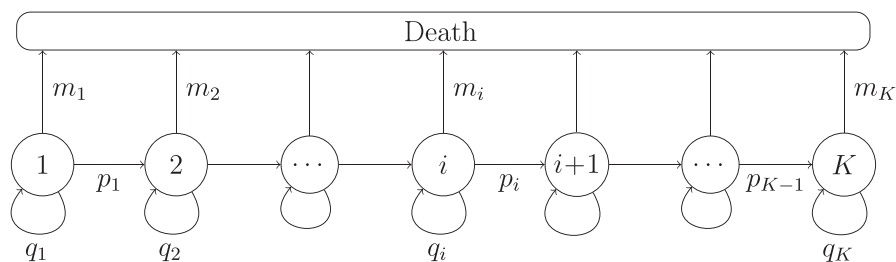


FIGURE 1 Life cycle for a matrix growth model of forest dynamics. Circles represent the diameter classes. Arrows represent the possible transitions between two consecutive time steps. Between time t and $t + 1$, a tree in diameter class i dies with probability m_i , stays alive in class i with probability q_i or moves up from class i to $i + 1$ with probability p_i .

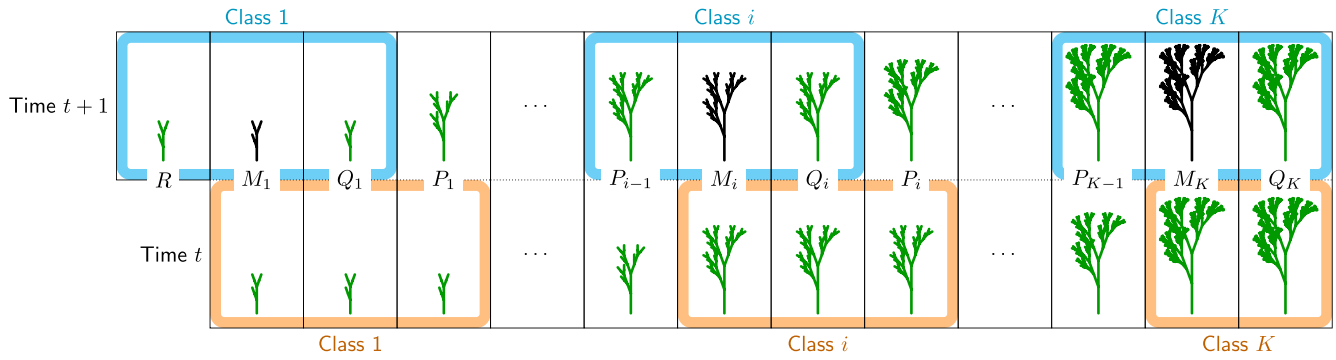


FIGURE 2 Given a dataset with the observed status (alive or dead) and diameter of N trees at times t and $t + 1$, representation of the count of trees in each diameter class at each time step according to their status. Diameter classes are represented by coloured boxes: orange = diameter classes for the forward transitions; blue = diameter classes for the backward transitions. For each diameter class, $i = 1, \dots, K$, the counts of tree are: M_i , counts of trees that die in diameter class i between time t and $t + 1$; Q_i , count of trees that stay alive in diameter class i ; P_i , count of trees that move up from diameter class i to $i + 1$. In the first class, R is the number of recruited trees between time t and $t + 1$.

Markov chains is commonly done using Bayes' formula (Surace & Scandi, 2023):

$$\Pr(X_t = i | X_{t+1} = j) = \frac{\Pr(X_{t+1} = j | X_t = i) \Pr(X_t = i)}{\sum_k \Pr(X_{t+1} = j | X_t = k) \Pr(X_t = k)}$$

These calculations are continued in Appendix A. However, this approach has an inherent limitation in defining RMM because the number of past dead individuals in a population is unknown. To compute the number of dead individuals in matrix models, it is enough to know the number of living individuals in each state. However, to compute the backward transitions from death, the cumulative number of dead individuals must be known. In other words, we can define stasis and reverse growth as backward transitions, but we cannot define reverse mortality as a backward transition by solely monitoring living individuals.

Nonetheless, RMM can be defined in a way similar to matrix models by counting the individuals that experience backward transitions and expressing these numbers in relative terms to the number of living individuals in each state. For backward transitions, the number of living trees in diameter class i at time $t + 1$ is: $O_i = Q_i + P_{i-1}$ for each class $i = 2, \dots, K$ (Figure 2). The stasis rate, that is the rate of trees that were already in class i at time t , is: $r_i = (M_i + Q_i) / O_i$. The reverse growth rate, that is the rate of trees in diameter class i at time $t + 1$ that were issued from diameter class $i - 1$, is: $b_i = P_{i-1} / O_i$. Contrary to the forward transition rates that sum to one, the sum of backward transition rates is greater than one: $r_i + b_i = 1 + M_i / O_i$. The quantity $r_i + b_i - 1$ now represents the mortality rate in class i . The first diameter class is a bit different from the others because it does not have any reverse growth rate but includes the newly recruited trees. These recruited trees have to be removed to compute the stasis rate, which is $r_1 = (M_1 + Q_1) / (O_1 - R) = 1 + M_1 / Q_1$, where $O_1 = Q_1 + R$ is the number of living trees in diameter class 1 at time $t + 1$.

The backward matrix relationship thus is $\mathbf{N} = \mathbf{C}(\mathbf{O} - \mathbf{R})$, where the reverse growth transition matrix $\mathbf{C}$ is

$$\mathbf{C} = \begin{bmatrix} r_1 & b_2 & & 0 \\ & r_2 & \ddots & \\ & & \ddots & b_K \\ 0 & & & r_K \end{bmatrix} \quad (2)$$

As in matrix models, recruitment in RMM can also be expressed proportionally to $\mathbf{O}$ rather than in an additive form. The backward matrix relationship then is $\mathbf{N} = \mathbf{V}\mathbf{O}$, where the reverse transition matrix $\mathbf{V}$ is

$$\mathbf{V} = \begin{bmatrix} r_1 - h & b_2 - h & -h & \dots & -h \\ & r_2 & b_3 & & 0 \\ & & \ddots & \ddots & \\ & & & \ddots & b_K \\ 0 & & & & r_K \end{bmatrix} \quad (3)$$

and h is the average fecundity rate of the reverse matrix model. Given the counts of individuals in diameter classes, the expression of h is $h = (1 + M_1 / Q_1) R / (O_1 + \dots + O_K)$.

Given a dataset with the observed diameters of trees at time t and $t + 1$, including the diameter of trees that were alive at time t but dead at $t + 1$ and the diameter of trees that were recruited between t and $t + 1$, the expressions of r_i , b_i and h allow us to fit RMM to observations. The fitted backward recurrence relationship can then be iterated to retrodict past dynamics.

2.3 | Inversion of the recurrence relationship

When a forward matrix model is already fitted but the goal is to retrodict past dynamics rather than predict future dynamics, the question is to invert the recurrence relation of the forward model. We considered four methods to invert the forward recurrence relation. One was based on RMM and provided interpretability of the

inversion process in terms of vital rates. The other three inversion methods aimed to solve for the input term of the recurrence without focusing on the biological interpretation of the inversion process: one was a numerical method without an explicit expression of the solution, while the other two were based on zero- and first-order approximations of the recurrence relation, allowing for an explicit expression of the solution to be computed. When the forward transition matrix has fixed coefficients, these three methods boil down to matrix inversion. Differences among the three methods arise in the more complicate case of density-dependent matrix models.

2.3.1 | Numerical inversion

Given a vector of abundances $\mathbf{O}$, a density-dependent matrix model can be numerically inverted to find the inverse image $\mathbf{N}$ of $\mathbf{O}$. It consists in using a numerical algorithm to solve the optimization problem $\min_{\mathbf{N}} \|\mathbf{O} - \mathbf{G}(\mathbf{N})\mathbf{N} - \mathbf{R}(\mathbf{N})\|$, where $\|\cdot\|$ is a norm of $\mathbb{R}^K$ (e.g. the ℓ^2 -norm). Numerical optimization methods require a starting point for $\mathbf{N}$. Any of the solutions presented in the subsequent sections can be used as a starting point for the numerical optimization method.

2.3.2 | Zero-order approximation

A zero-order approximation of the inverse image of the vector of abundances $\mathbf{O}$ was obtained by replacing $\mathbf{N}$ by $\mathbf{O}$ in the expressions of the transition matrix and recruitment vector. In other words, instead of solving the non-linear equation $\mathbf{O} = \mathbf{G}(\mathbf{N})\mathbf{N} + \mathbf{R}(\mathbf{N})$ for $\mathbf{N}$, we solve the linear equation $\mathbf{O} = \mathbf{G}(\mathbf{O})\mathbf{N} + \mathbf{R}(\mathbf{O})$. This approximation makes sense if $\mathbf{N}$ is close enough to $\mathbf{O}$, typically if the time step of the matrix model is small enough (e.g. 1 year) so that small changes are observed between times t and $t + 1$. Because the determinant of $\mathbf{G}$ is $|\mathbf{G}| = q_1 q_2 \dots q_K > 0$, matrix $\mathbf{G}$ is invertible. Therefore, the zero-order approximation of the back-projected state is

$$\mathbf{N} = \mathbf{G}(\mathbf{O})^{-1} [\mathbf{O} - \mathbf{R}(\mathbf{O})] \quad (4)$$

If the first element of $\mathbf{R}$ is a linear function of $\mathbf{N}$, the zero-order approximation is rather obtained by solving $\mathbf{O} = \mathbf{U}(\mathbf{O})\mathbf{N}$, where $\mathbf{U}$ is a Usher transition matrix. Under mild conditions (see [Supporting Information S2](#)), we can show that an Usher matrix is invertible. The zero-order approximation of the back-projected state thus is $\mathbf{N} = \mathbf{U}(\mathbf{O})^{-1}\mathbf{O}$.

2.3.3 | First-order approximation

A first-order approximation of the inverse image of the vector of abundance $\mathbf{O}$ was obtained by making a first-order Taylor expansion at $\mathbf{O}$ of the application ϕ that maps $\mathbf{N}$ onto $\mathbf{G}(\mathbf{N})\mathbf{N} + \mathbf{R}(\mathbf{N})$. The first-order expansion of ϕ at $\mathbf{O}$ is (Magnus & Neudecker, 2007): $\phi(\mathbf{O}) + \mathbf{D}_{\phi}(\mathbf{O})(\mathbf{N} - \mathbf{O})$, where $\mathbf{D}_{\phi}$ is the Jacobian matrix of ϕ , that is the

$K \times K$ matrix whose j th element is the partial derivative of the i th element of $\phi(\mathbf{N})$ with respect to the j th element of $\mathbf{N}$. Using this first-order expansion, an approximate solution of $\mathbf{O} = \phi(\mathbf{N})$ thus is

$$\mathbf{N} = \mathbf{O} + \mathbf{D}_{\phi}(\mathbf{O})^{-1} [\mathbf{O} - \mathbf{G}(\mathbf{O})\mathbf{O} - \mathbf{R}(\mathbf{O})] \quad (5)$$

Using the chain rule of matrix differentiation, $\mathbf{D}_{\phi}(\mathbf{N}) = (\mathbf{N} \otimes \mathbf{I}_K) \mathbf{J}(\mathbf{N}) + \mathbf{G}(\mathbf{N}) + \mathbf{D}(\mathbf{N})$, where $\mathbf{J}$ is the Jacobian matrix of $\mathbf{G}$, $\mathbf{D}$ is the Jacobian matrix of $\mathbf{R}$, $\mathbf{I}_K$ is the $K \times K$ identity matrix, and $\otimes$ denotes the Kronecker product (Magnus & Neudecker, 2007, p. 31).

We can readily check that [Equation \(5\)](#) generalizes [Equation \(4\)](#). Indeed, the Jacobian matrix of the application that maps $\mathbf{N}$ onto $\mathbf{G}(\mathbf{O})\mathbf{N} + \mathbf{R}(\mathbf{O})$ is $\mathbf{G}(\mathbf{O})$. Replacing $\mathbf{D}_{\phi}(\mathbf{O})$ by $\mathbf{G}(\mathbf{O})$ in [Equation \(5\)](#) brings back [Equation \(4\)](#).

2.3.4 | RMM as an inversion method

By replacing M_i , Q_i and P_i by $m_i N_i$, $q_i N_i$ and $p_i N_i$ respectively in the expressions of r_i and b_i , one can express the backward transition rates as a function of the forward transition rates. It brings

$$\begin{cases} r_1 = (1 - p_1)/q_1 \\ r_i = (1 - p_i)/(q_i + p_{i-1}/\eta_i) & \text{for } i = 2, \dots, K-1 \\ b_i = p_{i-1}/(p_{i-1} + q_i \eta_i) & \text{for } i = 2, \dots, K \\ r_K = 1/(q_K + p_{K-1}/\eta_K) \end{cases} \quad (6)$$

where $\eta_i = N_i / N_{i-1}$. The expression of b_i given here is exactly the one obtained by the retrodiction of Markov chains (see [Equation 7](#) in [Appendix A](#)), whereas the expression of r_i slightly differs from that obtained in this manner (see [Appendix A](#)). Using these expressions, a reverse matrix model can straightforwardly be built from a given matrix model. In other words, RMM can also be seen as an inversion method. Even when their transition rates are related by [Equation \(6\)](#), the backward transition matrix $\mathbf{C}$ is not the inverse of the forward matrix $\mathbf{G}$ ([Supporting Information S4](#)). Nonetheless, $\mathbf{GC}$ and $\mathbf{CG}$ are close to the identity matrix.

2.4 | Case studies

Two case studies illustrate RMM and the inversion of matrix models. First, we fit a reverse matrix model to a dataset from a tropical rainforest and compare the retrodicted dynamics to past observations. Second, we inverse a density-dependent matrix model and compare the capacity of the four inversion methods to match the predictions of the forward model.

2.4.1 | RMM of a tropical rain forest

We used the data from the Paracou experimental site in French Guiana to fit a reverse matrix model (Gourlet-Fleury et al., 2004).

In 1984, twelve 6.25-ha permanent plots were set up at Paracou. Each plot was divided in four 1.5625-ha subplots, which are here the repetition units. All trees with diameter at breast height greater than 10 cm were inventoried. The plots have been monitored since 1984, on an annual basis until 1995, then every 2 years. Diameters have been remeasured, and dead trees and newly recruited trees have been recorded at each survey. The 12 plots were distributed in three blocks. In each block, three silvicultural treatments of increasing intensity, combining selective logging and additional thinning, were applied, each to one plot. The fourth plot in each block was left as control. Thus, for each treatment, there were twelve 1.5625-ha repetitions. Treatments were applied between 1986 and 1988, then no intervention was made and natural dynamics occurred. Treatments removed on average 18.2% (treatment 1), 36.1% (treatment 2) and 43.9% (treatment 3) of the basal area of 1986 (Gourlet-Fleury et al., 2004, p. 256). In 2003, the forest on the treated plots was still recovering from the disturbance that had occurred 17 years earlier.

We fitted a RMM for each treatment separately. We used RMMs with fixed transition rates and recruitment proportional to $\mathbf{O}$, that is with the transition matrix given by Equation (3). Following Favrichon (1998), we used 11 diameter classes ranging from 10 to 60 cm dbh with a constant width of 5 cm, except the last class that gathered all trees greater than 60 cm dbh. Observations in 2001 and 2003 were used to fit the RMMs. Therefore, the time step of the RMMs was 2 years. Forest dynamics were then retrodicted till 1985.

2.4.2 | Inversion of a density-dependent matrix model

We back-projected the density-dependent matrix model developed by Lin et al. (1996) to study the forest tree demography and financial income of the northern hardwoods in the United States. This model relies on $K = 12$ diameter classes. The time step is 1 year. There are three species groups, denoted I to III. Upgrowth rates depend on the plot basal area: $p_{gi} = \beta_{gi} + \beta_g B$, where p_{gi} is the upgrowth rate for trees in species group g and diameter class i , B is the plot basal area, and β_{gi} and β_g are fixed coefficients. The plot basal area is computed as $B = \sum_{g=1}^{\text{III}} \mathbf{t} \mathbf{B} \mathbf{N}_g$, where $\mathbf{B}$ is the vector giving the average basal area of trees in each diameter class and $\mathbf{N}_g$ is the vector of tree abundances per diameter class for species group g . Mortality rates m_{gi} are density-independent. Stasis rates are computed as $q_{gi} = 1 - m_{gi} - p_{gi}$ (where, by convention, $p_{gK} = 0$) and are thus also density-dependent. The recruitment vector for species group g is $\mathbf{R}_g = [R_g, 0, \dots, 0]$, where the recruitment rate is also density-dependent, $R_g = \alpha_{g0} + \alpha_{g1} B + \alpha_{g2} N_g$, where N_g is the total abundance of species group g and the α_{gk} s are fixed coefficients. The total abundance of species group g is computed as $N_g = \mathbf{1}_K \mathbf{N}_g$, where $\mathbf{1}_K$ is the vector of length K full of ones. In total, the model has 30 parameters, including 9 for recruitment, 12 for growth and 9 for mortality.

Starting from an empty plot, we projected the state of the forest using Lin et al.'s density-dependent matrix model over a time horizon of T , bringing a final vector $\mathbf{N}(T)$ of tree abundances in the three species groups. Starting from $\mathbf{N}(T)$, we then back-projected

the state of the forest for the same period of time T . We used the four back-projection techniques described in the previous sections: numerical inversion, zero-order approximation, first-order approximation and RMM. For a given horizon T , let $\mathbf{N}(t)$ and $\mathbf{O}(t)$ be the vectors of tree abundances at time $t = 1, \dots, T$ when moving forward and backward, respectively, with $\mathbf{N}(T) = \mathbf{O}(T)$. We defined divergence between forward projection $\mathbf{N}(t)$ and backward projection $\mathbf{O}(t)$ as an absolute difference in basal area of any of the species group greater than $0.1 \text{ m}^2 \text{ ha}^{-1}$ and a relative difference in basal area of any of the species group greater than 5%. To assess divergence, we also computed the root mean square error (RMSE) of the difference between $\mathbf{N}(t)$ and $\mathbf{O}(t)$. The distance to stable structure was defined as the RMSE of the difference between $\mathbf{N}(t)$ and $\lim_{t \rightarrow \infty} \mathbf{N}(t)$. We defined divergence time t_{div} as the maximum value of t such that $\mathbf{O}(t)$ diverges from $\mathbf{N}(t)$. When moving backward, t_{div} actually gives the year when divergence starts. We defined the duration of agreement between the forward and backward projections as $\tau = T - t_{\text{div}}$. Finally, we defined the elasticity of the duration of agreement to a model parameter θ as the relative rate of change of the duration of agreement with respect to the relative rate of change of θ , that is $\partial \ln \tau / \partial \ln \theta$.

3 | RESULTS

3.1 | Retrodicted dynamics at Paracou

The reverse growth rate increased from control plots to the plots under treatment 3, following the trend in perturbation intensity (Table 1, Supporting Information S5). So did the fecundity rate. The stasis rate showed the opposite trend. To summarize, the more intense the treatment, the faster the reverse dynamics.

RMM extrapolated the dynamics observed in 2001–2003 to the past. For those plots that were not disturbed (control plots) or slightly disturbed in 1986–1988 (treatment 1), the dynamics were quite linear and the RMM retrodiction conformed to observed dynamics (Figure 3a,b and Figure S1a,b in Supporting Information S6). However, for more intense disturbance (treatments 2 and 3), the post-disturbance plot dynamics showed non-linearity. The RMM with fixed transition rates was then

TABLE 1 Transition rates and dominant eigenvalue of the reverse transition matrices fitted to forest plots at Paracou, French Guiana.

Treatment	$\bar{b}$	h	$\bar{r}$	λ
T0	0.089	0.021	0.939	0.998
T1	0.109	0.042	0.918	0.988
T2	0.152	0.049	0.880	0.971
T3	0.169	0.061	0.863	0.966

Note: These plots underwent silvicultural treatments of increasing intensity, from T0 (control plots) to T3. $\bar{b}$ is the average reverse growth rate across diameter classes, h is the fecundity rate, $\bar{r}$ is the average stasis rate across diameter classes, and λ is the dominant eigenvalue.

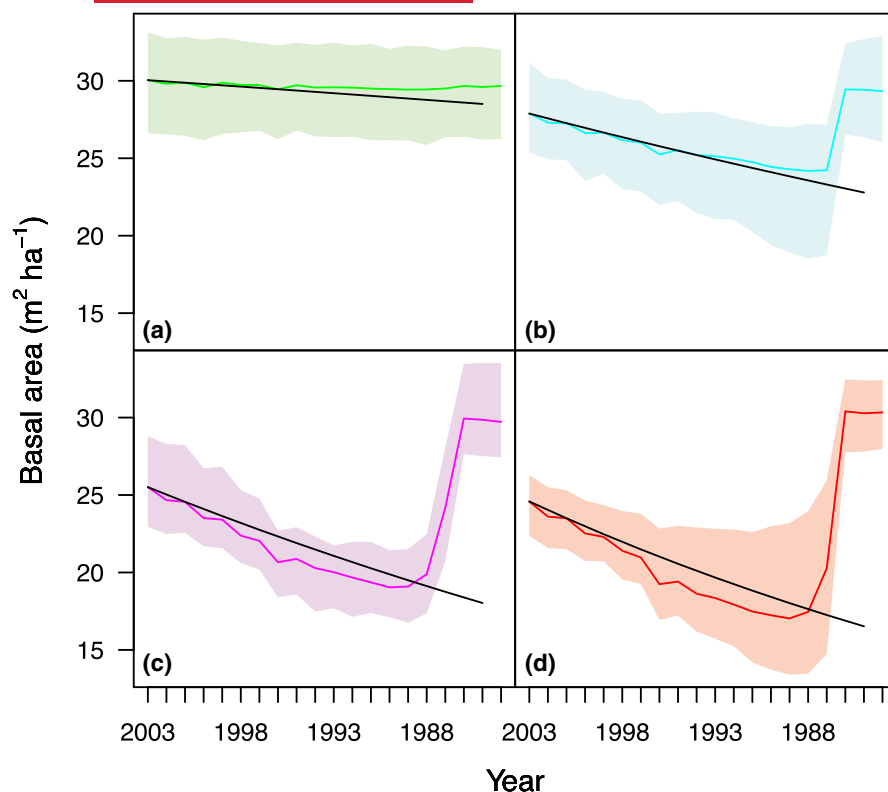


FIGURE 3 The retrodicted basal area of the rainforest at Paracou, French Guiana, matched observations from 2003 back to 1988. Retrodiction was achieved using a reverse matrix model fitted to the observations of 2001 and 2003. The black line is the model retrodiction. The coloured line is the observed average trajectory and the shaded area is the observed envelope of the trajectories of twelve 1.5625-ha plots. These plots underwent silvicultural treatments between 1986 and 1988: (a) control plots (no treatment), (b) treatment 1 (low disturbance intensity), (c) treatment 2 (medium disturbance intensity) and (d) treatment 3 (high disturbance intensity).

less successful at retrodicting dynamics. The agreement between retrodictions and observations was acceptable for basal area (Figure 3c,d) but showed limitations for the density of trees (Figure S1c,d in Supporting Information S6).

As for forward matrix models, the long-term behaviour of RMM at Paracou was determined by the dominant eigenvalue of the reverse transition matrix $\mathbf{V}$. The dominant eigenvalue decreased from control plots to the plots under treatment 3, following the opposite trend of the perturbation intensity (Table 1). All eigenvalues were less than one. Therefore, moving backwards from 2003, RMM predicted a long-term origin of the forest that was a void forest. However, this predicted backward decline (= recovery in the chronological order) was slow (almost at equilibrium) for controls plots. To summarize, the more intense the treatment, the faster the long-term backward decline.

3.2 | Inversion of Lin et al.'s model

In Lin et al.'s density-dependent matrix model, the growth transition matrix $\mathbf{G}$ and the recruitment vector $\mathbf{R}$ linearly depend on $\mathbf{N}$, so the application ϕ that maps $\mathbf{N}$ onto $\mathbf{G}(\mathbf{N})\mathbf{N} + \mathbf{R}(\mathbf{N})$ depends quadratically on $\mathbf{N}$. Therefore, the Jacobian $\mathbf{J}$ of $\mathbf{G}$ and the Jacobian $\mathbf{D}$ of $\mathbf{R}$ are constant, whereas the Jacobian $\mathbf{D}_\phi$ of ϕ is linear with respect to $\mathbf{N}$. The computations of the Jacobian matrices for Lin et al.'s matrix model are given in Supporting Information S3.

Starting from an empty plot, Lin et al.'s matrix projection model predicted a rapid increase in basal area (till ≈ 50 years), followed by slow dampened oscillations that finally converged (after ≈ 500 years)

to a stable diameter structure (see black lines in Figure 4 and Supporting Information S7). Starting from different initial diameter structures, the same stable structure was reached (Figure 5 and Supporting Information S8). The dampened oscillations of basal area that are observed over time correspond to a spiral pattern in the phase space (Figure 5d).

Starting from an empty plot, the backward projections using the first-order approximation succeeded to follow the forward trajectories over at least 100 years (Figure 4, Figures S2–S4 and Table 2). In particular, when projecting the diameter distributions from $T = 100$ years backwards, the backward model succeeded to go back to the initial state of an empty plot. When initiating the retrodiction from a state closer to the stable structure, the inversion method was able to follow the spiral of the phase space for a bit less than a cycle, then quickly diverged from it.

The numerical inversion method for backward projections had the same performance as the first-order approximation (Figure 4, Figures S2–S4). Backward projections with the numerical inversion method diverged from the forward projections at the same time as with the first-order approximation (Table 2). However, the computing time was 20 times longer with the numerical inversion method than with the first-order approximation.

The backward projections using the zero-order approximation succeeded to follow the forward trajectories over a shorter period of time than with the first-order approximation (Figure 4, Figures S2–S4). Accordingly, the divergence time with this method was closer to the time horizon of the forward projections (Table 2). The backward projections using RMM was able to follow the forward trajectories over an even shorter period of time (Figure 4,

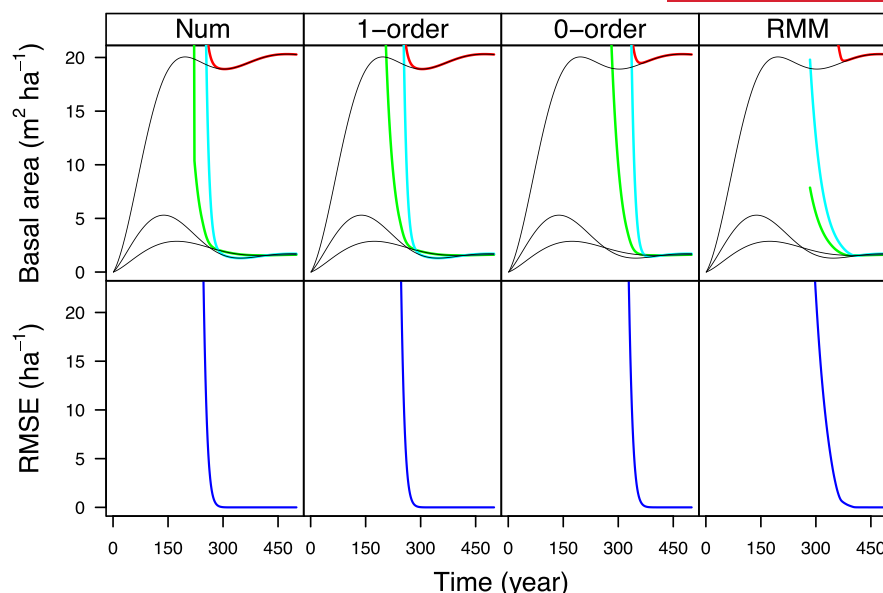
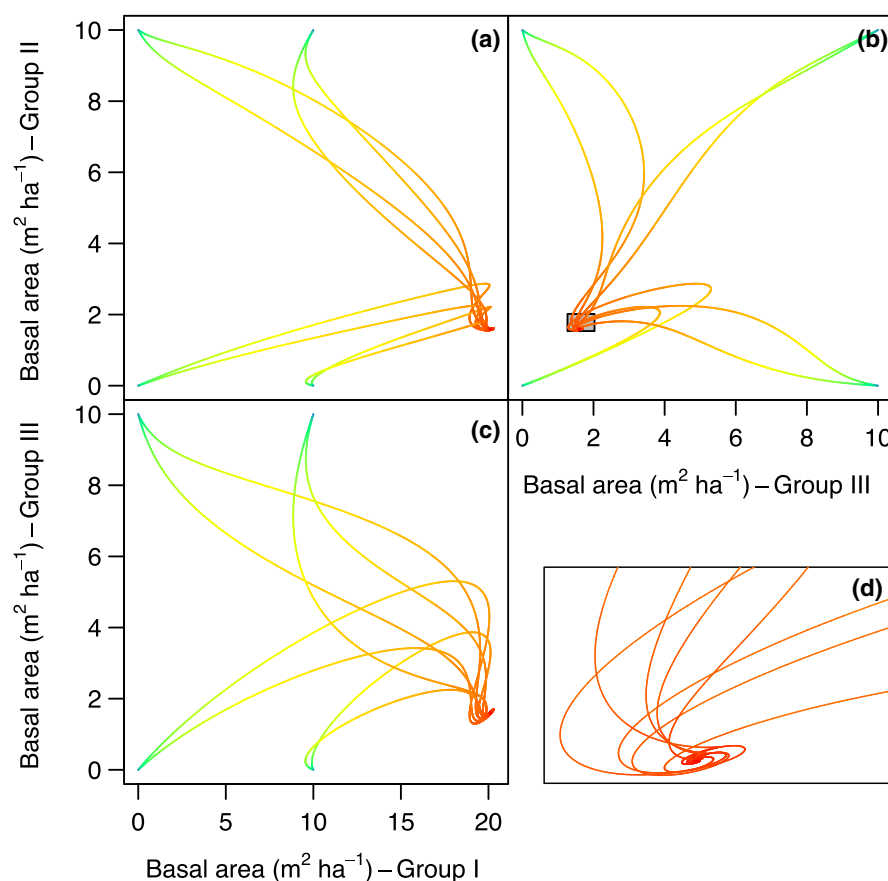


FIGURE 4 After 500 years of forward prediction using the density-dependent matrix model by Lin et al. (1996), and starting from the resulting state, backward projections closely tracked the predicted basal area for 62 to 207 years (depending on the inversion method) before quickly diverging. Top row shows the forward (black lines) and backward (coloured lines) projections of basal area; the three lines correspond to the three species groups of Lin et al. (1996). Bottom row shows the root mean square error of the retrodicted diameter structure, where the error is calculated with respect to the structure predicted by the forward model. Backward projections are made using four methods (in columns): Numerical inversion ("num"), first-order approximation ("1-order"), zero-order approximation ("0-order") and reverse matrix model ("RMM").

FIGURE 5 In the forward projection, starting from different initial states, the density-dependent model by Lin et al. (1996) converged to the same equilibrium point. To avoid overloading the figure, only eight different initial states are shown. Projections are shown in the three-dimensional subspace of the phase space corresponding to the basal areas of the three species groups: (a) basal area of group II versus basal area of group I; (b) basal area of group III versus basal area of group I; (c) basal area of group III versus basal area of group I. Panel (d) is a zoom of the grey area shown in panel (b). Colours correspond to the time steps (early time is shown in green and late time in red).



Figures S2–S4). Its divergence time was closer to the time horizon of the forward projections than with the other inversion methods (Table 2).

Starting from an initial empty plot, the duration of agreement τ between the forward and backward projections had a tendency to increase as the time horizon T of the forward projection increased (Figure 6). Nonetheless, the duration of agreement also depended on the specific position of the state reached at time T with respect to the stable structure. This dependence of τ on the position resulted in a succession of peaks and troughs in the τ versus T curve, separated by a duration equal to one cycle of the spiral of the phase space (Figure 6a). When following a forward trajectory, the successive position changed in a continuous way, so the τ versus T curve also varied continuously. When plotting τ against the distance to the stable structure, this was no longer true (Figure 6b). The succession of peaks and troughs along the cycles then resulted in an hysteresis pattern.

Whatever the inversion method used, the duration of agreement τ was less sensitive to the recruitment parameters of the matrix model than to its growth or mortality parameters (Figure 7). RMM was the least sensitive inversion method to the parameters of the matrix model, while the zero-order approximation was the most sensitive one.

TABLE 2 Divergence time between the forward projections of the density-dependent matrix model by Lin et al. (1996) and the backward projections, depending on the method used for model inversion and the time horizon of forward projections. An empty cell means that there was no divergence between the forward and backward projections.

Inversion method	Time horizon T (year)			
	100	300	500	700
Numerical inversion	—	147	293	505
First-order approximation	—	147	293	505
Zero-order approximation	21	200	377	585
Reverse matrix model	50	227	438	640

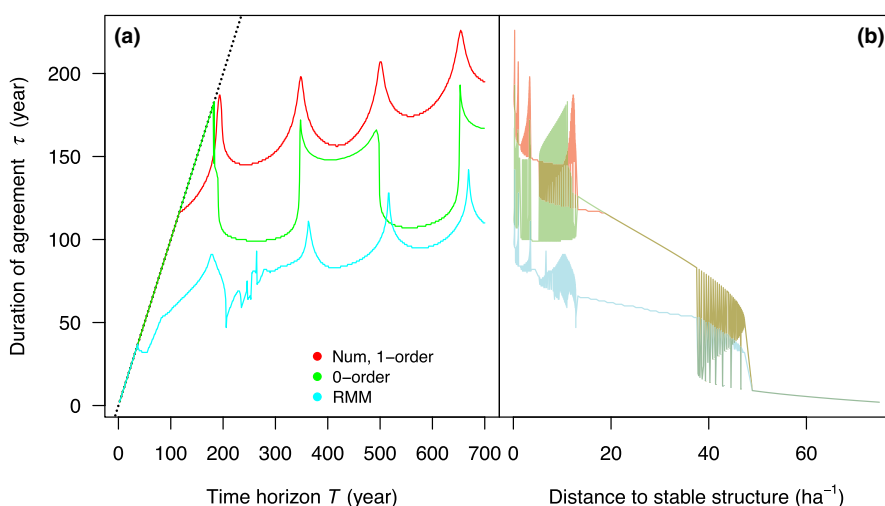


FIGURE 6 The duration of agreement τ between the forward and backward projections of the density-dependent matrix model by Lin et al. (1996) (a) non-linearly depended on the time horizon T of the forward projection and (b) non-uniquely depended on the distance to the stable diameter structure. Backward projections are made using four methods: Numerical inversion (red line), first-order approximation (red line too), zero-order approximation (green line) and reverse matrix model (cyan line). In (a), the dotted line is the line $y = x$.

4 | DISCUSSION

Retrodiction allows the matrix models, a modelling approach that has proven effective in predicting future population dynamics, to now be applied for exploring and understanding past dynamics. To retrodict population dynamics using the matrix modelling framework, one has two options: use a reverse matrix model or inverse a forward matrix model. The latter option can be preferred when a matrix model is already available. Inversion methods can thus benefit from the large pool of existing matrix models (Salguero-Gómez et al., 2015, 2016). In contrast, when no forward model a priori exists and observations are to be used to build a model, the former option can be preferred. In both cases, one must remain aware of the functional assumptions made in any modelling work and of the influence that the model choice can have on predictions (Bell et al., 2013; Rueda-Cediel et al., 2015). Even when a forward matrix model already exists, RMM can be seen as a way to re-assess these assumptions.

4.1 | Retrodiction using RMM

4.1.1 | Defining RMM

We defined RMM using the counts of individuals in diameter classes and the proportions relative to the number of living individuals at time $t + 1$. This approach brought the same expression of the reverse growth rate as the retrodiction of the Markov chain based on the life cycle. Markov chain retrodiction tells us that we may end up with little capacity to reconstruct the past if too much information is lost in the forward direction (Ellison et al., 2009; Rupprecht & Vural, 2018). Because matrix modelling loses the information on dead trees in the forward direction, we can anticipate that the perfect retrodiction of matrix models will not be possible. Nonetheless, matrix models depart from the Markov chain formalism in several aspects, so that results from Markov chain retrodiction cannot be directly applied to matrix modelling.

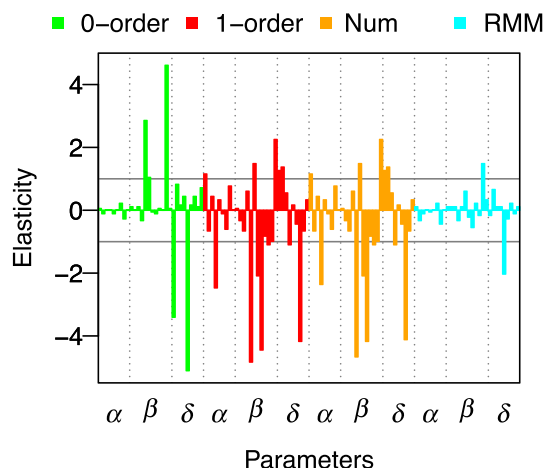


FIGURE 7 Elasticities of the duration of agreement to the parameters of the density-dependent matrix model by Lin et al. (1996). The duration of agreement is computed using four inversion methods: Numerical inversion (orange), first-order approximation (red line), zero-order approximation (green) and reverse matrix model (cyan). The matrix model has 30 parameters that can be related to three dynamics processes: α -parameters (9 parameters) relate to recruitment, β -parameters (12) relate to growth, and δ -parameters (9) relate to mortality.

Partial differential equations provide another standpoint on matrix model retrodiction. A matrix model can be seen indeed as a finite difference approximation of a partial differential equation (Caswell, 2001, Chap. 8; Takada & Hara, 1994). If the solution of the partial differential equation exists and is unique, it predicts the size structure of the population at any time, whether in the past or in the future (Ackleh et al., 2007). Nevertheless, because the matrix model is a dissipative finite difference scheme (Picard & Liang, 2014), this mathematical standpoint also let us expect some inherent limitation in reconstructing the past from current situation.

4.1.2 | Possible applications of RMM

While this study primarily focuses on case studies with specific matrix models in the context of forest dynamics, the conceptual framework of RMM and its adaptability suggest broader applicability across diverse ecosystems. Our objective was to define the structure of the transition matrix of RMM and the proportion estimators of its transition rates. We reckon that this initial step will make possible a diverse range of methodological developments, paralleling those of matrix modelling (Caswell, 2001).

The ability of the model to retrodict forest demography can be instrumental in ecosystem management (Tester et al., 1997). Effective ecosystem management requires to understand ecosystem dynamics over long time horizons, including past historical dynamics (Beller et al., 2020). RMM can make the link between past, current and future population dynamics, thus contributing to historical ecology. RMM can specifically be employed to assess the success

of forest restoration initiatives (Melichar et al., 2022). By comparing retrodicted and observed outcomes, practitioners can refine and adapt restoration strategies, ensuring that efforts align with pre-designated ecological goals.

RMM's ability to project backward in time will also be useful for biodiversity conservation (Heppell et al., 2000). Conservationists can combine retrodictions and projections to assess the impact of management practices on different species and evaluate the efficacy of conservation interventions over extended periods (Crone et al., 2013). Populations may be structured by age or stage rather than size, leading to Leslie or Lefkovich transition matrices instead of Usher matrices. The same reasoning used to define RMM for size-structured populations applies to age- or stage-structured populations. We also reckon that specific matrix modelling methods developed for species-rich ecosystems, like tropical rainforests or coral reefs, will be transferable to RMM (Mortier et al., 2015).

Specifically, the intrinsic rate of increase of a population given by the dominant eigenvalue λ of the transition matrix is a key instrument in population viability analysis (Morris & Doak, 2002, Chap. 7). In the case study at Paracou, the long-term retrodicted dynamics of the population was also determined by the dominant eigenvalue of the reverse transition matrix. $\lambda < 1$ in RMM corresponds to a growing population in the chronological order. However, the properties of λ in matrix modelling result from the Perron-Frobenius theorem that stands for non-negative matrices. Because the reverse transition matrix $\mathbf{V}$ has negative terms, this theorem no longer applies to RMM. Further mathematical research will be needed to relate the dominant eigenvalue of $\mathbf{V}$ to the long-term behaviour of RMM and develop life table response experiments for RMM (Caswell, 1989).

Furthermore, RMM will be useful to assess global change impact. Given the dynamic nature of ecosystems and the influence of external factors such as climate change and land-use/land-cover change (Claeys et al., 2019), RMM provides a valuable tool for retroactively assessing the impacts on forest dynamics. It can also assist in reconstructing past trajectories of forest carbon stocks, thereby enhancing our understanding of the current and future capacity of forests to mitigate climate change (Gasser et al., 2020; Magerl et al., 2019). By incorporating relevant environmental variables into the model as it has been done for matrix models (Liang et al., 2011), RMM can contribute to retrodicting how climate change has altered forest composition, structure and overall health.

In forestry, the application of RMM can aid in optimizing timber harvesting strategies and implement sustainable and economically viable management (Lin et al., 1996). The case study at Paracou illustrated how to fit a RMM to data to retrodict dynamics triggered by logging. The nonlinear post-logging dynamics and the dependence of the transition matrices on disturbance intensity suggested that a density-dependent RMM may be more appropriate than a RMM with fixed coefficients. This conclusion is consistent with Favrichon (1998) who fitted a density-dependent forward matrix model at Paracou. The same techniques used in matrix

modelling to introduce density-dependence can be transferred to RMM (Caswell, 2001, Chap. 16; Solomon et al., 1986).

RMM's capability to retrodict past states and identify potential divergence points enhances its utility in risk assessment (Charles et al., 2009). Forest managers can use this information to evaluate resilience plans, preparing for and mitigating the impact of disturbances such as pests, diseases, or extreme weather events (Martinez-Ramos & Cristi  n, 1997). Environmental stochasticity has been introduced in matrix modelling to account for natural disturbances, using matrix resampling, parametric matrix distribution, or stochastic shocks (Dalglish et al., 2010; Davison et al., 2010). We reckon that the same techniques will be transferable to RMM.

Finally, the transparency of the RMM modelling process and its visualization through forward and backward projections make it an excellent tool for educational purposes. Communicating complex ecological concepts to various stakeholders, including policymakers and the general public, can foster a broader understanding of the importance of sustainable forestry and conservation practices (Arrontes, 2021; Mugasha et al., 2017).

4.2 | Inversion of predictions

When a forward matrix model is available, retrodiction can be achieved by inverting the function that maps the state at time t to the state at time $t + 1$. Inversion methods are thus analytical methods in the mathematical sense of the term. However, with the exception of RMM that offers interpretable parameters, they tell nothing about the backward transitions. The various inversion methods matched the predictions more or less well. Nonetheless, the numerical error of the inversion should not be confused with the prediction error of the model (Diele & Marangi, 2019; Petrovskii & Petrovskaya, 2012). If a forward matrix model has a poor predictive performance, matching it closely with a good inversion method will still bring a poor retrodiction. For that reason, we still recommend RMM as an inversion method even if it diverged sooner from the forward dynamics than the other inversion methods.

Inversion methods succeeded to retro-project the forest dynamics predicted by Lin et al.'s matrix model over several decades. These results will have to be confirmed with other matrix models. We reckon that these inversion methods will remain efficient as long as the matrix model is not too much nonlinear (i.e. if the ϕ function that maps $\mathbf{N}$ onto $\mathbf{O}$ is smooth enough). Machine-learning methods (such as random-forest) are increasingly being used to fit density-dependent relationships in forest dynamics models and bring greater nonlinearity in those relationships (Liu et al., 2018; Ma et al., 2020).

When the starting point of the backward projection gets closer to the stationary state of the matrix model, the back-projection gets less and less able to retrodict when the past dynamics was still transient. As different starting points converge to the same steady state, knowing the ending point (i.e. the steady state) brings no information

about the starting point (Rupprecht & Vural, 2018). In other words, moving towards a stationary state results in the loss of some information and then there is an inherent limit to retrodiction due to the information lost on the way forward (Ellison et al., 2009; Rupprecht & Vural, 2018). Practically, before doing retrodiction, it may be useful to map the attractors of a matrix models in its phase space as we did in Figure 5, so as to check where the starting point of back projection is located with respect to the steady state of the model.

When using RMM as an inversion method, the backward transition matrix obtained is not exactly the inverse of the forward transition matrix. The product of $\mathbf{G}$ and $\mathbf{C}$ rather is a tridiagonal matrix (Supporting Information S4). Takada and Hara (1994) showed that such tridiagonal transition can be interpreted as the transition matrix of a diffusion process. Consider the i th size class that contain a given number of individuals. If $\mathbf{C}$ was the inverse of $\mathbf{G}$, computing one step forward and one step backward would result in all individuals staying in class i . With the tridiagonal matrix, most of the individuals are still in class i but a small proportion of them has also spread to classes $i - 1$ and $i + 1$. Diffusion terms in transition matrix can be introduced to account for growth variability (Takada & Hara, 1994). It remains to be seen if the off-diagonal terms in $\mathbf{GC}$ and $\mathbf{CG}$ are consistent with individual growth variability.

5 | CONCLUSIONS

RMM is an approach to retrodict the matrix model used in ecology to project population dynamics. Its transition rates are similar to the transition probabilities of the retrodicted Markov chain associated to the life cycle of the population dynamics, with a major difference as regards mortality. RMM was able to retrodict the dynamics of a disturbed tropical rainforest and back-project the dynamics predicted by Lin et al.'s density-dependent matrix model over a few decades. Inversion methods based on the inversion of the recurrence relationship of the matrix model were able to back-project predictions over a longer period of time. Because a model is a simplified representation of reality, when having data to fit a model without knowing the true forest dynamics, it makes more sense to fit a RMM than to fit a matrix model and inverse it.

AUTHOR CONTRIBUTIONS

All authors conceived the ideas. Nicolas Picard and Jack Andrew Hansen designed the methods and analysed the data. Nicolas Picard and Jingjing Liang led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declares no conflict of interests.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210X.14426>.

DATA AVAILABILITY STATEMENT

The code used in this paper can be found in Zenodo via <https://zenodo.org/doi/10.5281/zenodo.13351330> (Picard, 2024). The Paracou data are available at Guyafor, DataBase of the French Guyana Permanent Plot Network, Cirad-CNRS-ONF (<https://paracou.cirad.fr/website/resources>).

ORCID

Nicolas Picard  <https://orcid.org/0000-0001-5548-9171>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Supporting Information S1. List of symbols.

Supporting Information S2. Invertibility of a Usher transition matrix.

Supporting Information S3. Jacobian of the density-dependent matrix model of Lin et al. (1996).

Supporting Information S4. Product of a forward and backward transition matrix.

Supporting Information S5. Backward transition matrices at Paracou.

Supporting Information S6. Retrodicted forest dynamics at Paracou, French Guiana.

Supporting Information S7. Inversion of Lin et al.'s density-dependent matrix model.

Supporting Information S8. Interactive three-dimensional view of the projections of the model by Lin et al. (1996).

Supporting Information S9. Comments on the R code.

Appendix A. RMM using retrodiction of Markov chains.

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